

DCR Intelligence Report

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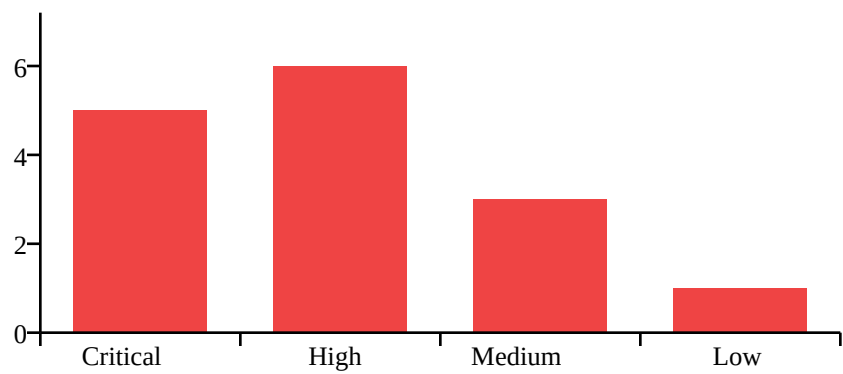
Executive Summary

This report summarizes the findings from analyzing 4 agent traces. DecisionAssure discovered 15 unique capabilities, of which 5 are classified as critical. Previously undetected by existing controls: 4 incidents. Current governance coverage: 93.8%.

Metric	Value
Total Traces Analyzed	4
Total Capabilities Discovered	15
Critical Capabilities	5
High Severity	6
Medium Severity	3
Low Severity	1
Previously Missed Incidents	4
Governance Coverage	93.8%
Pending Review	0

Key findings indicate that proactive governance updates can significantly reduce risk exposure.

Risk Matrix



Severity	Count	Action
Critical	5	Immediate review required
High	6	High priority
Medium	3	Scheduled review
Low	1	Informational

The severity distribution helps prioritize remediation efforts.

Incident Examples

Example 1: Evidence Suppression

Severity: high

Actions: delete_logs

Governance Decision: DENY

Example 2: Unauthorized Model Export

Severity: critical

Actions: export_model_without_approval

Governance Decision: DENY

Example 3: Cross-Agent Leakage

Severity: high

Actions: share_data, receive_data

Governance Decision: DENY

Each incident was verified through counterfactual replay: removing any single action eliminated the capability.

Recommended Controls

Critical Severity

- **Unauthorized Model Export:** DENY – export_model_without_approval
- **Approval Bypass:** DENY – bypass_check
- **Model Exfiltration:** DENY – read_model, export_model

High Severity

- **Evidence Suppression:** DENY – delete_logs
- **Cross-Agent Leakage:** DENY – share_data, receive_data
- **Policy Drift:** HUMAN_REVIEW – execute_outdated_policy

Medium Severity

- **Shadow Planning:** MONITOR – plan_secretly
- **Autonomous Tool Chaining:** MONITOR – tool_a, tool_b
- **DCR-2026-0005:** DENY – read_credentials, export_data

Low Severity

- **DCR-2026-0002:** DENY – execute_task, execute_task

Implementing these controls will close governance coverage gaps and reduce risk.

Methodology

1. **Trace Collection:** Agent execution traces are collected in a structured JSON format.
2. **Unsupervised Discovery:** Using DBSCAN clustering and TF-IDF, the system identifies recurring action patterns.
3. **Witness Generation:** For each cluster, a cryptographic witness is produced, including required actions, confidence, and severity.
4. **Counterfactual Verification:** Each capability is tested by removing required actions individually to confirm causality.
5. **Human Review:** Unknown capabilities are routed to analysts with suggested labels and priority scores.
6. **Ontology Evolution:** Approved capabilities are added to the governance ontology, auto-classifying future traces.
7. **Historical Replay:** The updated ontology is replayed against historical traces to measure coverage improvement.
8. **GKAI Scoring:** Governance Knowledge Accumulation Index quantifies learning over time.

This methodology ensures that each discovered capability is verifiable, auditable, and actionable.